The Torsional Markov Memory-Offload Pump:
A Three-Scale Quaternionic Delta-Chain with Inverse Compensation and Non-Equilibrium Circulation

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June 21, 2026

Abstract

We formalize a three-scale quaternionic Markov pump whose rotational memory is distributed across three coupled cyclic gears: an outer high-resolution memory layer (12-point gear, 720-frame chain), a middle inverse-compensating layer (10-point gear, 72-frame chain), and an inner coarse-equilibrium layer (8-point gear, 10-frame chain). Each layer stores unit-quaternion increments (deltas); coarse layers are obtained from fine layers by ordered products of consecutive deltas, and fine detail is reconstructed from coarse layers by fractional-power (SLERP) lifts. We show that the block products telescope to exact net rotations, that the variable down-sampling blocks partition the frame index, and that the fractional-power lift reconstructs exactly on the principal branch. Treating the three gear layers as the states of a Markov chain, a net cyclic current $J_{12,10} + J_{10,8} + J_{8,12} > 0$ encodes directed (non-reversible) circulation of memory rather than passive diffusion; at stationarity the three edge currents coincide, so the pump condition reduces to positivity of a single cycle current. Geometric coherence is maintained by closing the cycle product $H = Q^{(8)}Q^{(10)}Q^{(12)}$ to the identity up to a controlled holonomy, with residual drift redistributed across the gears. We close by describing how this multi-resolution memory hierarchy interoperates with the executable, parity-safe Hyperbolic Sparsemax Pump, with which it shares both the unit-quaternion state space and the broken-detailed-balance current condition.

1 Introduction

The system is a three-scale quaternionic Markov pump whose memory is distributed across three coupled cyclic gears. Informally,

$$\text{large memory chain} \longleftrightarrow \text{compressed inverse correction chain} \longleftrightarrow \text{coarse equilibrium chain},$$

where information flows both upward and downward through quaternionic delta maps while preserving rotational coherence. The outer chain is the high-resolution memory layer, the middle chain is the inverse compensating layer, and the inner chain is the coarse torsional anchor. The remainder of this paper makes each arrow precise: downward flow compresses fine rotation history into coarse torsional updates, and upward flow lifts coarse rotations back into finer chains through quaternion interpolation. A three-state Markov current supplies the directed circulation that distinguishes an active memory pump from passive reversible storage, and a holonomy constraint keeps the whole mechanism in geometric equilibrium.

2 The Three Quaternion Chains

Definition 1 (Gear states)**.** Let the three gears carry unit-quaternion states

$$q_n^{(12)} \in \mathbb{H}\ (n = 0, \ldots, 719), \qquad q_m^{(10)} \in \mathbb{H}\ (m = 0, \ldots, 71), \qquad q_\ell^{(8)} \in \mathbb{H}\ (\ell = 0, \ldots, 9),$$

with

$$\left\| q_n^{(12)} \right\| = \left\| q_m^{(10)} \right\| = \left\| q_\ell^{(8)} \right\| = 1.$$

The outer chain (12-point gear, 720 frames) stores the full non-compressed rotation history; the middle chain (10-point gear, 72 frames) stores the inverse compensating correction; and the inner chain (8-point gear, 10 frames) stores the coarse equilibrium structure.

3 Outer 720-Frame Chain

The outer gear stores the rotation history as quaternionic deltas,

$$\Delta q_n^{(12)} = q_n^{(12)} \left(q_{n-1}^{(12)}\right)^{-1},$$

so that the full state is reconstructed by the ordered product

$$q_n^{(12)} = \Delta q_n^{(12)}\, \Delta q_{n-1}^{(12)} \cdots \Delta q_1^{(12)}\, q_0^{(12)}.$$

With 720 frames distributed over 12 gear positions there are $720/12 = 60$ frames per outer tooth. Writing the tooth index and the local frame-within-tooth index as

$$a(n) = n \bmod 12, \qquad r(n) = \left\lfloor \tfrac{n}{12} \right\rfloor,$$

yields the outer memory address $q_{a,r}^{(12)}$ with $a \in \mathbb{Z}_{12}$ and $r \in \mathbb{Z}_{60}$.

4 Middle 72-Frame Inverse Chain

The middle gear runs inversely, so its update direction is opposite that of the outer gear. Its inverse delta is written

$$\Delta q_m^{(10)} = \left(q_{m-1}^{(10)}\right)\left(q_m^{(10)}\right)^{-1}$$

rather than $q_m^{(10)}(q_{m-1}^{(10)})^{-1}$; in this orientation the middle chain absorbs and cancels torsional drift from the outer layer. With 72 frames over 10 positions the natural scale is

$$\tfrac{72}{10} = 7.2$$

frames per middle tooth. Because this is not an integer, the middle gear is phase-shearing relative to the outer and inner gears, which prevents the same alignment from recurring too quickly. The middle address and phase are

$$b(m) = m \bmod 10, \qquad \varphi_{10}(m) = \tfrac{m}{72},$$

and the inverse chain is reconstructed (in left-acting form) by

$$q_m^{(10)} = \left(\Delta q_m^{(10)}\, \Delta q_{m-1}^{(10)} \cdots \Delta q_1^{(10)}\right)^{-1} q_0^{(10)}.$$

5 Inner 10-Frame Chain on the 8-Point Gear

The lowest gear is the coarsest memory layer, with deltas

$$\Delta q_\ell^{(8)} = q_\ell^{(8)} \left(q_{\ell-1}^{(8)}\right)^{-1}.$$

With 10 frames running on an 8-point gear, the phase and global phase are

$$c(\ell) = \ell \bmod 8, \qquad \varphi_8(\ell) = \tfrac{\ell}{10}.$$

The mismatch between 10 frames and 8 gear points produces a small torsional fold each cycle, the innermost folding step.

6 Inter-Scale Coupling

The system scales in one consistent direction, $720 \to 72 \to 10$. The down-sampling (projection) maps are

$$P_{12\to10}(n) = \left\lfloor \tfrac{n}{10} \right\rfloor, \qquad P_{10\to8}(m) = \left\lfloor \tfrac{10m}{72} \right\rfloor,$$

and the corresponding lifting maps are

$$L_{10\to12}(m) = 10m, \qquad L_{8\to10}(\ell) = \left\lceil \tfrac{72\ell}{10} \right\rceil.$$

These let information flow down,

$$q^{(12)} \xrightarrow{P_{12\to10}} q^{(10)} \xrightarrow{P_{10\to8}} q^{(8)},$$

and back up,

$$q^{(8)} \xrightarrow{L_{8\to10}} q^{(10)} \xrightarrow{L_{10\to12}} q^{(12)}.$$

This is the memory-offloading pipeline.

7 Downward Offload I: Outer to Middle

To offload the outer 720-frame chain into the middle 72-frame chain, group every ten consecutive outer deltas into one coarse update,

$$\Delta Q_m^{(12\to10)} = \Delta q_{10m+9}^{(12)}\, \Delta q_{10m+8}^{(12)} \cdots \Delta q_{10m}^{(12)}, \qquad m = 0, \ldots, 71,$$

and store its inverse in the middle gear,

$$\Delta q_m^{(10)} = \big(\Delta Q_m^{(12\to 10)}\big)^{-1},$$

so that the middle layer acts as a stabilizing counter-rotation. The coarse update is exactly the net rotation across its block.

Proposition 2 (Block telescoping)**.** *For every m,*

$$\Delta Q_m^{(12\to 10)} = q_{10m+9}^{(12)} \big(q_{10m-1}^{(12)}\big)^{-1}.$$

Proof. Each factor is $\Delta q_k^{(12)} = q_k^{(12)}(q_{k-1}^{(12)})^{-1}$. In the ordered product the inner factors cancel pairwise:

$$\Delta q_{10m+9}^{(12)} \cdots \Delta q_{10m}^{(12)} = q_{10m+9}^{(12)}\big(q_{10m+8}^{(12)}\big)^{-1} q_{10m+8}^{(12)}\big(q_{10m+7}^{(12)}\big)^{-1} \cdots q_{10m}^{(12)}\big(q_{10m-1}^{(12)}\big)^{-1},$$

leaving $q_{10m+9}^{(12)}(q_{10m-1}^{(12)})^{-1}$. (Associativity of quaternion multiplication is all that is used; commutativity is not.) □

This realizes the first offload: 720 fine frames become 72 inverse blocks.

8 Downward Offload II: Middle to Inner

Compressing the 72 middle deltas into 10 inner deltas requires variable block sizes, since $72/10 = 7.2$. For each inner index ℓ define the block

$$A_\ell = \Big\{ m : \ \big\lfloor \tfrac{10m}{72} \big\rfloor = \ell \Big\},$$

and set the coarse update to the chronologically ordered product

$$\Delta Q_\ell^{(10\to 8)} = \prod_{m\in A_\ell}^{\leftarrow} \Delta q_m^{(10)}.$$

The order matters because quaternion multiplication is noncommutative. The inner delta is either this product or, when the inner gear is to act as an equilibrium anchor, its normalization,

$$\Delta q_\ell^{(8)} = \Delta Q_\ell^{(10\to 8)} \qquad \text{or} \qquad \Delta q_\ell^{(8)} = \mathrm{Normalize}\big(\Delta Q_\ell^{(10\to 8)}\big).$$

Proposition 3 (The blocks partition the middle index)**.** *The sets $A_0, \dots, A_9$ are disjoint and cover $\{0, \dots, 71\}$, with $\sum_{\ell=0}^{9} |A_\ell| = 72$. Each block has size $|A_\ell| \in \{7, 8\}$.*

Proof. The map $m \mapsto \lfloor 10m/72 \rfloor$ is nondecreasing and sends $\{0, \dots, 71\}$ onto $\{0, \dots, 9\}$ (at $m = 71$, $\lfloor 710/72 \rfloor = 9$), so its level sets A_ℓ partition the domain. Each level set is an interval of integers of length either $\lceil 7.2 \rceil = 8$ or $\lfloor 7.2 \rfloor = 7$, and the lengths sum to the size of the domain, 72. □

This realizes the second offload: 72 inverse frames become 10 coarse equilibrium frames.

9 Upward Reconstruction

To reconstruct fine detail, coarse rotations are lifted by fractional powers. For unit quaternions the fractional power is defined through the exponential map,

$$q^w = \exp\big(w \log q\big),$$

which is the standard spherical-linear (SLERP) interpolation of a rotation. From inner to middle, with $\ell(m) = \lfloor 10m/72 \rfloor$ and a fractional weight w_m,

$$\Delta_c q_m^{(10)} = \big(\Delta q_{\ell(m)}^{(8)}\big)^{w_m}, \qquad \widehat{q}_m^{(10)} = \Delta_c q_m^{(10)}\, \widehat{q}_{m-1}^{(10)};$$

and from middle to outer, with $m(n) = \lfloor n/10 \rfloor$,

$$\Delta_c q_n^{(12)} = \big(\Delta q_{m(n)}^{(10)}\big)^{1/10}, \qquad \widehat{q}_n^{(12)} = \Delta_c q_n^{(12)}\, \widehat{q}_{n-1}^{(12)}.$$

Upward flow thus proceeds $10 \to 72 \to 720$. The $1/10$ exponent distributes one coarse rotation evenly across the ten fine frames it governs.

Proposition 4 (Exact fractional reconstruction)**.** *Let $q \in \mathbb{H}$ be a unit quaternion with rotation angle in $(-\pi, \pi)$, and take the principal branch of* log*. Then $\big(q^{1/N}\big)^N = q$ for every positive integer N; in particular ten applications of $\Delta_c q_n^{(12)} = (\Delta q_{m(n)}^{(10)})^{1/10}$ recover the coarse increment $\Delta q_{m(n)}^{(10)}$.*

Proof. On the principal branch $\log q = \frac{\theta}{2}\,\hat{\boldsymbol{u}}$ for a unit imaginary axis $\hat{\boldsymbol{u}}$ and angle $\theta \in (-\pi, \pi)$, so $q^{1/N} = \exp\big(\frac{1}{N} \log q\big)$ has axis $\hat{\boldsymbol{u}}$ and angle θ/N with $|\theta/N| < \pi$. Powers of a quaternion about a fixed axis add angles, hence $\big(q^{1/N}\big)^N$ has axis $\hat{\boldsymbol{u}}$ and angle θ, i.e. equals q. □

10 Markov Pump Structure

Treat each gear layer as the state of a Markov chain, $X_t \in \{12, 10, 8\}$, where 12 is the high-resolution outer memory, 10 the inverse middle correction, and 8 the coarse inner equilibrium. Let

$$K = \begin{pmatrix} K_{12,12} & K_{12,10} & K_{12,8} \\ K_{10,12} & K_{10,10} & K_{10,8} \\ K_{8,12} & K_{8,10} & K_{8,8} \end{pmatrix}, \qquad \pi = (\pi_{12},\, \pi_{10},\, \pi_8),$$

be a row-stochastic transition matrix with stationary distribution π. The directed memory currents on the three-cycle are

$$J_{12,10} = \pi_{12} K_{12,10} - \pi_{10} K_{10,12}, \qquad J_{10,8} = \pi_{10} K_{10,8} - \pi_8 K_{8,10},$$

$$J_{8,12} = \pi_8 K_{8,12} - \pi_{12} K_{12,8}.$$

The torsional pump condition is a net directed circulation,

$$J_{12,10} + J_{10,8} + J_{8,12} > 0,$$

meaning the system actively circulates memory rather than diffusing reversibly.

Lemma 5 (The cycle current is single-valued)**.** *For an irreducible chain on* $\{12, 10, 8\}$ *with stationary* π *(so* $\pi K = \pi$*), the three edge currents coincide,*

$$J_{12,10} = J_{10,8} = J_{8,12} =: J,$$

and hence the pump condition $J_{12,10} + J_{10,8} + J_{8,12} > 0$ *is equivalent to* $J > 0$*.*

Proof. Stationarity at state 12 gives $\pi_{12} = \sum_j \pi_j K_{j,12}$, i.e. $\pi_{12}(K_{12,10}+K_{12,8}) = \pi_{10}K_{10,12}+\pi_8 K_{8,12}$ after cancelling the self-term $\pi_{12}K_{12,12}$ from both sides. Rearranged, this is $J_{12,10} = J_{8,12}$. The same balance at state 10 gives $J_{10,8} = J_{12,10}$. Thus all three currents are equal, and their sum is $3J$. □

Remark 6. On a single cycle there is exactly one independent current, so the three-term condition is, at stationarity, the positivity of that one circulation; Lemma 5 makes this explicit. This is the discrete analogue of a nonzero Schnakenberg cycle affinity [3].

11 Equilibrium and Holonomy Control

To keep the mechanism in geometric equilibrium, require the net cycle product to close to the identity up to a small, controlled holonomy. Writing the per-layer net rotations

$$Q^{(12)} = \prod_{n=0}^{719} \Delta q_n^{(12)}, \qquad Q^{(10)} = \prod_{m=0}^{71} \Delta q_m^{(10)}, \qquad Q^{(8)} = \prod_{\ell=0}^{9} \Delta q_\ell^{(8)},$$

form the holonomy

$$H = Q^{(8)} Q^{(10)} Q^{(12)}.$$

Perfect closure is $H = 1$, and in practice one enforces $\|\log H\| \approx 0$. After each cycle the states are renormalized, $q \leftarrow q/\,\|q\|$, the residual drift $\varepsilon = \log H$ is measured, and the correction is distributed across the gears,

$$\Delta q_k^{(12)} \leftarrow \exp(-\lambda_{12}\,\varepsilon)\,\Delta q_k^{(12)}, \quad \Delta q_k^{(10)} \leftarrow \exp(-\lambda_{10}\,\varepsilon)\,\Delta q_k^{(10)}, \quad \Delta q_k^{(8)} \leftarrow \exp(-\lambda_8\,\varepsilon)\,\Delta q_k^{(8)},$$

with the allocation weights summing to one,

$$\lambda_{12} + \lambda_{10} + \lambda_8 = 1.$$

This is a structure-preserving (Lie-group) renormalization that keeps the pump coherent across cycles [4].

12 Inward Folding (Schematic)

The inward folding across cycles can be described schematically by a recursive scale map. Starting from the outer layer and alternating the projection direction,

$$F_0 = q^{(12)}, \qquad F_1 = P_{12\to 10}(F_0)^{-1}, \qquad F_2 = P_{10\to 8}(F_1),$$

each pump cycle updates by

$$F_{r+1} = P_r(F_r)^{\sigma_r}, \qquad \sigma_r \in \{+1, -1\}.$$

The middle gear carries $\sigma_{10} = -1$ because it runs inversely, while the outer and inner gears carry $\sigma_{12} = \sigma_8 = +1$, giving the sign pattern $(+, -, +)$. This produces inward folding without losing orientation coherence.

Remark 7 (Reading of this section). The equations here are a schematic for the alternating, sign-staggered folding; $P_{12\to 10}$ and $P_{10\to 8}$ denote the layer down-sampling acting on the corresponding chain, whose precise quaternionic form is the ordered-product offload of Sections 7–8, and σ_r records the inverse orientation of the middle gear. The sign pattern $(+, -, +)$ is the operative content.

13 Algorithm and Compact Form

The full memory-offloading procedure is:

(1) Store outer deltas $\Delta q_n^{(12)} = q_n^{(12)}(q_{n-1}^{(12)})^{-1}$ for every frame n.

(2) Compress outer blocks into middle inverse deltas, $\Delta q_m^{(10)} = \big(\Delta q_{10m+9}^{(12)} \cdots \Delta q_{10m}^{(12)}\big)^{-1}$ for $m = 0, \ldots, 71$.

(3) Compress middle deltas into inner coarse deltas, $\Delta q_\ell^{(8)} = \mathrm{Normalize}\big(\prod_{m\in A_\ell}^{\leftarrow} \Delta q_m^{(10)}\big)$ for $\ell = 0, \ldots, 9$.

(4) Compute the Markov current $J_{ij} = \pi_i K_{ij} - \pi_j K_{ji}$ and require $J_{12,10} + J_{10,8} + J_{8,12} > 0$.

(5) Check cycle closure $H = Q^{(8)}Q^{(10)}Q^{(12)}$ and enforce $\|\log H\| \approx 0$.

(6) Renormalize every quaternion, $q \leftarrow q/\,\|q\|$.

(7) Flow data upward when needed via $\Delta_c q = \exp(w \log q)$ to lift coarse rotations into finer chains.

Compactly, the mechanism is the multi-resolution delta chain

$$q_n^{(12)} = \Big(\prod_{r=n}^{1} \Delta q_r^{(12)}\Big) q_0^{(12)},$$

with offload maps

$$\Delta q_m^{(10)} = \Big(\prod_{r=10m+9}^{10m} \Delta q_r^{(12)}\Big)^{-1}, \qquad \Delta q_\ell^{(8)} = \mathrm{Normalize}\Big(\prod_{m\in A_\ell}^{\leftarrow} \Delta q_m^{(10)}\Big),$$

subject to the Markov pump condition $J_{12,10} + J_{10,8} + J_{8,12} > 0$ and the quaternionic equilibrium condition $H = Q^{(8)}Q^{(10)}Q^{(12)} \approx 1$. The outer layer stores fine memory, the middle layer stores inverse torsional correction, and the inner layer stores coarse equilibrium structure.

14 Interoperability with the Hyperbolic Sparsemax Pump

The companion Hyperbolic Sparsemax Pump [1] is an executable, parity-safe transition kernel on the 64-voxel SHD–CCP lattice: rational, row-stochastic, exactly sparse, and free of transcendental operations. The two constructions are complementary rather than redundant, and they share enough structure to be coupled directly.

A common non-equilibrium primitive. Both pumps are driven by the same broken-detailed-balance current. The sparsemax pump defines a base-referenced current $J^0_{ij} = \pi^0_i \widehat{K}_{ij} - \pi^0_j \widehat{K}_{ji}$ on the lattice; the present pump defines $J_{ij} = \pi_i K_{ij} - \pi_j K_{ji}$ on the gear layers. These are the same primitive at two scales: the sparsemax current circulates orientation *within* the fine lattice, while the torsional current (Lemma 5) circulates memory *between* the three coarse layers. A single design quantity, the cycle affinity, governs both.

A shared state space. Both layers represent states as unit quaternions. The sparsemax voxel spinors $q_i = (w_i, \boldsymbol{m}_i) \in S^3 \cap \mathbb{Q}^4$ are exactly the kind of unit quaternion stored by the outer chain as $q_n^{(12)}$. The sparsemax kernel can therefore act as the per-frame generator of the outer memory chain: each lattice transition emits a unit-quaternion increment, and the offload pump compresses the resulting 720-frame history downward and reconstructs it upward on demand.

A clean rational/transcendental split. The sparsemax forward path is deliberately exact-rational, with no exp, log, or root, so that the discrete parity invariant is bit-reproducible. The offload pump's upward reconstruction, by contrast, uses fractional powers $q^w = \exp(w \log q)$, and its holonomy correction uses $\varepsilon = \log H$. This division of labour maps precisely onto the sparsemax paper's octet-residual quarantine, which requires that transcendental and irrational quantities never re-enter the rational kernel: one keeps the rational sparsemax kernel on the executable forward path and confines the SLERP lifts and holonomy correction to the offline memory-reconstruction layer, where such residuals are already quarantined. Where reconstructed states must re-enter the lattice, they are rationalized first, exactly as the quarantine prescribes.

A shared torsional index. Both constructions carry a 12-fold torsional structure: the sparsemax lattice resolves its face-neighbor transitions into 12 torsional coupling channels, and the outer gear here is a 12-point gear. Identifying the two gives one 12-fold index that the fine kernel scores and the coarse pump compresses.

In combination, the sparsemax pump is the rational, parity-safe transition engine at the leaf scale, and the torsional offload pump is the multi-resolution memory hierarchy that stores, compresses, and reconstructs the orientation history that engine produces, both governed by the same directed-circulation condition. Together they realize a complete context-compression loop for the SHD–CCP setting that both papers address.

15 Scope and Outlook

This paper specifies a constructive model. The local results are proved: the block products telescope to exact net rotations (Proposition 2), the down-sampling blocks partition the middle index (Proposition 3), the fractional-power lift reconstructs exactly on the principal branch (Proposition 4), and the three gear currents coincide at stationarity (Lemma 5). The global claims, that the pump offloads and reconstructs memory *coherently*, are an engineering design rather than a

theorem: quantitative reconstruction-error bounds (the fractional lift is exact only on a single principal-branch increment, not for accumulated drift) and the existence of an explicit transition matrix K and stationary π realizing a prescribed cycle affinity are natural next steps. The inward-folding recursion of Section 12 is schematic, as noted there. With these caveats, the mechanism is, mathematically, a multi-resolution quaternion delta chain with an inverse middle compensator and a non-equilibrium Markov circulation current, designed to interoperate with the executable Hyperbolic Sparsemax Pump.

References

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